

SAHABA: AI Muslim Companion Report

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1 Use Case and Goals

In Islam, the need for accurate guidance and information, such as the Quranic verses, prayer times, the direction of Qibla, and the date conversion of the Islamic calendar, is important for the correct conduct of day-to-day religious activities. Incorrect information can make worship and prayers invalid [14]. This problem is particularly critical in non-Muslim countries, as local religious authority may be absent and information may be scarce and scattered across different sources [6, 15].

To resolve this problem, we propose *Sahaba*, a multimodal Islamic assistant which combines different Islamic information services in a single interface and utilises text, audio, and visual modes to facilitate accurate and reliable information for the proper performance of daily religious activities.

The primary stakeholders of this system are Muslims in need of accurate religious guidance. The secondary stakeholders are developers and researchers who ensure the accuracy of the proposed solution and its compliance with Islamic principles. The target users require not only accurate religious information, but also for the system to be transparent, accessible, and easy to use [3].

2 System Design and Implementation

The architecture of this system is centered around a LangGraph ReAct agent [11, 19] and utilizes OpenAI's large language model (LLM), gpt-5-nano-2025-08-07. LangChain [10] was also used for multi-turn conversation state. The agent interacts with a set of REST APIs for retrieving authoritative content for Quranic verses in English/Arabic [1], prayer times, direction of Qibla, and Islamic calendar conversion [2], and geocoding [13]. The frontend of this architecture utilizes Streamlit [16] for building the interface, as shown in Figure 1. The interface is a single-page chatbot with a sidebar for configuration selection, location/date selection, audio, and a main chat interface.

Three configurations are available, each with progressively richer multimodalities:

- (1) **Configuration A:** Text-based chat only. Users may ask for information about Quranic verses, prayer times, Islamic-Gregorian calendar conversions, and other general queries about Islam.
- (2) **Configuration B:** Extends Configuration A with text-to-speech (TTS) output using OpenAI's gpt-4o-mini-tts with autoplay and replay capabilities.
- (3) **Configuration C:** includes all features of Configuration B, extended with voice-based input using an in-browser voice recorder (speech recognition using OpenAI's gpt-4o-mini-transcribe), visual Qibla compass, auto-location detection using GPS, and optional date context for queries.

Design decisions were made with consideration of Value Sensitive Design (VSD) [9] to ensure cultural and religious appropriateness to Islam, trust, and users' agency. The design prioritizes aspects of clarity and feedback; Arabic script is visually highlighted in a different colour than Latin script, and TTS and Qibla compass visuals offer redundant feedback. Human-AI interaction guidelines [4] were considered for multimodal design, accessibility, and error avoidance. For example, voice input is offered as an option with textual input. The persona and tone of the system

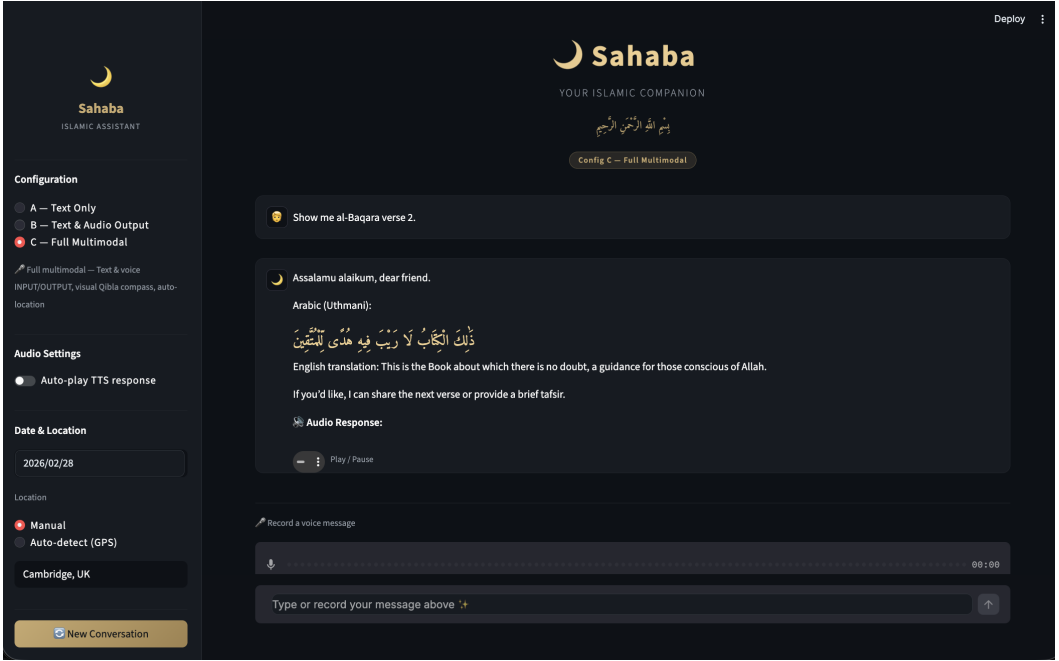


Fig. 1. Streamlit frontend interface of *Sahaba*.

are set to balance friendliness and clarity, incorporating respectful Islamic greetings with the aim of engaging users while being trustworthy.

3 Methods: Research Question, Evaluation Methods, and Metrics

3.1 Research Question

RQ1: How does increasing multimodality affect perceived trust, engagement, and usability in an Islamic AI assistant?

RQ2: What value tensions emerge as multimodality increases in a religious AI context?

3.2 Evaluation Methods and Metrics

To assess *Sahaba*, eight postgraduate Muslim students in non-Muslim-majority countries (5 females, 3 males; ages 20-30, $M=24.25$, $SD=3.03$) were recruited to represent users with limited access to religious information. Two evaluation strategies were employed to assess quantitative results and user experience [5].

Quantitative Evaluation. A comparative analysis of design variations for the three configurations was conducted. Four tasks were designed to represent user interactions:

- (1) Look up a Quranic verse
- (2) Find prayer times for a given location
- (3) Find Qibla direction
- (4) Ask a general Islamic question.

After each configuration, users were asked to complete a post-interaction survey to rate their experience on a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree) with the following questions:

- **Trust** [17]
 - (1) I trust the information provided by the assistant.
 - (2) I feel confident relying on this assistant for religious content.
- **Engagement** [12]
 - (1) I enjoyed interacting with the assistant.
 - (2) I was motivated to explore the assistant's features further.
- **Usability** (abbreviated System Usability Scale) [8, 18]
 - (1) I found the system easy to use.
 - (2) I think I would like to use this system frequently.
 - (3) I felt confident using the system.
 - (4) I thought the system was unnecessarily complex. (*reverse scored*)
 - (5) I found the system's features well integrated.

The scores were averaged for each configuration. Descriptive statistics and Friedman tests were used to determine significant differences.

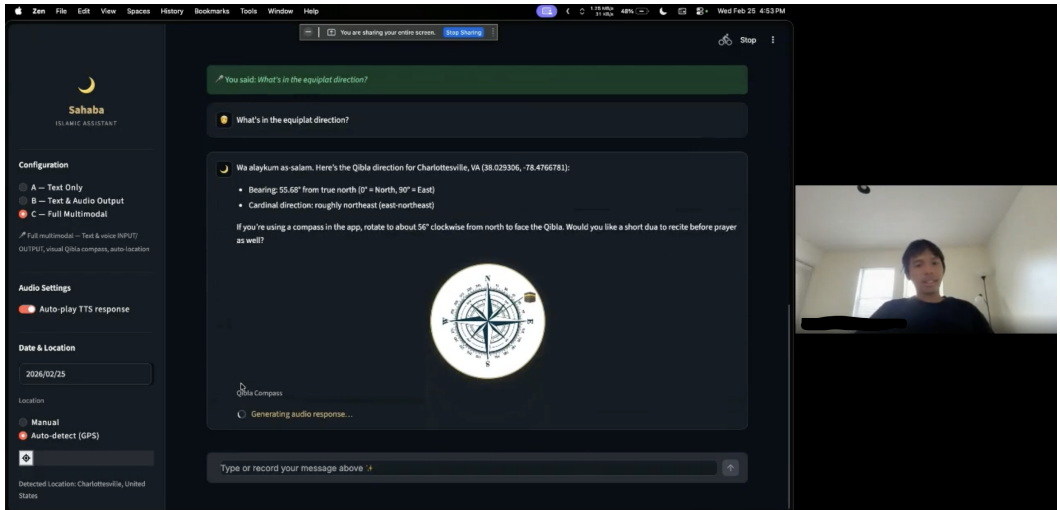


Fig. 2. A capture of a participant interacting with *Sahaba* during an evaluation session, verbalizing their thoughts.

Qualitative Evaluation. A combination of think-aloud sessions and post-interaction interviews was conducted. These techniques allowed us to understand the perceptions of participants on the religious appropriateness, trust, usability, accessibility, privacy, and engagement of the experience. During interviews, the following questions were asked:

- (1) How did it feel interacting with this configuration of the assistant?
- (2) Which features or modalities were most helpful?
- (3) Did anything feel confusing, distracting, or inappropriate?
- (4) Which configuration did you prefer and why? (*cross-comparison feedback*)

Participants’ responses were analysed using reflexive thematic analysis [7] to identify patterns, conflicts, and trade-offs that emerged from the multimodal experience.

Using this mixed-methods approach, we were not only able to measure the user-centric metrics but also examine the cultural, ethical, and modality-specific nuances of the system. Each session lasted 30-50 minutes. Figure 2 shows a participant using *Sahaba* during one of the sessions.

4 Results

Tables 1 and 2 summarize the quantitative analysis for the three configurations. **Trust**, **Engagement**, and **Usability** have small differences in mean scores between configurations, with configuration C showing moderately higher **Engagement** (6.38 ± 0.64) and **Usability** (6.17 ± 0.86). Moreover, Friedman tests revealed that effect sizes were small (Kendall’s $W \leq 0.244$), and that there were no statistically significant differences between configurations in all metrics (**Trust**: $\chi^2(2) = 3.90$, $p = 0.142$; **Engagement**: $\chi^2(2) = 1.65$, $p = 0.438$; **Usability**: $\chi^2(2) = 2.85$, $p = 0.241$).

Table 1. Descriptive Statistics for Trust, Engagement, and Usability (Mean $\pm$ SD; $N = 8$)

Metric	Configuration A	Configuration B	Configuration C
Trust	5.94 ± 1.74	5.00 ± 2.05	5.56 ± 1.24
Engagement	5.81 ± 1.31	5.81 ± 1.51	6.38 ± 0.64
Usability	5.90 ± 0.91	5.70 ± 1.35	6.17 ± 0.86

Table 2. Friedman Test Results for Trust, Engagement, and Usability ($N = 8$)

Metric	$\chi^2(2)$	p	Kendall’s W	Effect Size	Interpretation
Trust	3.90	0.142	0.244	Small	N.S.
Engagement	1.65	0.438	0.103	Small	N.S.
Usability	2.85	0.241	0.178	Small	N.S.

Qualitative analysis revealed four dominant themes. 6/8 participants emphasized Quranic content accuracy and tajweed, noting mispronunciations and Arabic text rendering errors that affected trust. 7/8 participants raised concerns about transparency issues as sourced information passed to the LLM becomes hallucinated. Multimodal experience and accessibility were appreciated by 7/8 participants, who found it easy to use features such as the TTS, voice input, and the Qibla compass, although some preferred text-only mode for its simplicity. The usability of the interface was a concern for 7/8 participants, citing the complexity of the sidebar controls and slow response time.

5 Discussion

Based on our analysis of *Sahaba*, several key value trade-offs and ethical issues in multimodal religious AI can be noted. While the use of multimodal features such as voice input, TTS, and Qibla compass helped in increasing the engagement and usability of the application for most participants, it also posed certain privacy issues, as users may not be comfortable in recording their sensitive religious queries. Similarly, the use of more modalities may increase the cognitive load on the user, especially those who prefer simple text-based interaction.

In a religious setting, trust and accuracy are of utmost importance. Participants emphasized how even slight changes in Quranic texts, such as differences in letters, vowels, punctuation marks, and

tajweed pronunciation, can cause differences in meaning. Meaning is extremely critical in Islam as the Quran is believed to be the literal word of God. Although *Sahaba* uses authoritative APIs, contents are passed through LLMs which may cause slight inaccuracies, again emphasizing the need to keep the content as it is before delivering it to users. Additionally, there are various school of thoughts (Madhhab) in Islam, and there are possibilities of variations in their interpretations. Although Quran's essence is consistent, variations in Hadith interpretations may cause differences in prayer time, method of worship and ablution, and other practices. A reliable assistant should take these variations into consideration and either explain them or let the user choose their own Madhhab.

Although *Sahaba* offers easy access to Islamic guidance and advice, participants identified the risk that users could accept the assistant's answers without verifying them. It is important to always cross-check critical religious guidance with qualified religious authorities and/or scholars. The assistant should instead be viewed as an aid to provide assistance and not the sole source of religious rulings to avoid taking religious decisions for granted.

In our study, the evaluation was restricted to a limited, homogeneous sample size ($N = 8$). Moreover, participants interacted with configurations without counterbalanced ordering. This might result in learning and fatigue effects, affecting the quality of rated scores. Future studies should explore methods for retaining the original form of the Quranic text and the hadith when using LLMs, support different madhhab variations, and conduct evaluation for larger, more diverse sample populations.

6 Conclusion

This study shows the effectiveness of multimodality in increasing engagement and usability, and facilitating trustworthy access to Islamic content. Participants valued accuracy, cultural suitability, and accessibility, but complexity and privacy issues arose with richer modalities. Multimodal approaches can improve Islamic AI experience if done transparently and with user control and respect for scriptural content.

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